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End Semester Examination December-2025

Name of the Program: B.Tech. Biotechnology
Name of the Course: Animal and Plant Biotechnology

Semester: V

Course Code: TBT-501

Time: 3Hours

Maximum Marks: 100

Note:-

- (v) All questions are compulsory.
- (vi) Answer any two sub questions among a, b & c in each main question
- (vii) Total marks in each main question are twenty.
- (viii) Each question carries 10 marks

Q1	(20marks)	
(a)	What is the principal of plant tissue culture? Explain how PTC works, and describe its advantages and limitations.	CO1
(b)	Relate the contributions of early researchers to modern-day tissue culture practices.	
(c)	Differentiate between dry heat, moist heat, and chemical sterilization methods in terms of mechanism and application.	
Q2	(20marks)	
(a)	Explain the process of somatic embryogenesis and its significance in plant biotechnology.	CO2
(b)	How are protoplast isolation and fusion performed? Discuss the advantages and limitations of this technique.	
(c)	Evaluate the advantages, limitations, and future prospects of synthetic seed technology in plant biotechnology.	
Q3	(20marks)	
(a)	Analyze the social, economic, and environmental impacts of the Green Revolution in India.	CO3
(b)	Evaluate the effectiveness of <i>Agrobacterium rhizogenes</i> in the genetic transformation of plants.	
(c)	What is Genetic transformation? Explain the direct and indirect methods of genetic transformation.	
Q4	(20marks)	
(a)	Highlight the contributions of any two pioneering scientists in the field of animal biotechnology, explaining how their discoveries or innovations advanced the development of animal cell culture.	CO4
(b)	Explain the concept of cell culture. Describe how cell cultures are established from animal tissues, and classify the different types of cell cultures based on their origin and characteristics.	
(c)	Write short notes on the following types of media used in animal cell culture: (1) Natural media, (2) Artificial media.	
Q5	(20marks)	
(a)	Describe the following methods used in the production of transgenic animals: (A) Retroviral vector method, (B) Sperm Mediated Gene Transfer.	CO5
(b)	Propose a detailed plan for implementing in-vitro fertilization (IVF) in a livestock improvement program in India.	
(c)	Critically evaluate the impact of modern biotechnological tools and techniques on livestock production.	

